

Progress Report 2: Food Macro Estimation from Cooking Videos

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1 Progress Since Report 1

Report 1 established the dataset pipeline (6 classes, 20 subclasses, 11,885 training images) and a baseline K-Means clustering approach on CLIP ViT-B/32 embeddings, achieving 98.35% class-level accuracy. Since then, the project has undergone a significant architectural upgrade and the full end-to-end pipeline is now operational.

- Upgraded encoder.** Replaced CLIP ViT-B/32 (512-dim) with **CLIP ViT-L/14** (OpenCLIP, LAION-2B pretrained, 768-dim embeddings). The larger encoder produces richer representations, improving downstream classification across all subclasses.
- Multi-label MLP classifier.** Replaced K-Means clustering with a trained two-layer MLP head ($768 \rightarrow 256 \rightarrow 20$) using **sigmoid activation + BCEWithLogitsLoss**. This enables multi-label detection (multiple ingredients per frame), which K-Means could not support. Per-class positive weighting handles class imbalance. The MLP achieves **99.2% macro F1** and **99.6% micro F1** on the test set, with all 20 subclasses exceeding 0.96 F1.
- Per-class threshold calibration.** Thresholds are tuned per-class on the validation set (sweep 0.75–0.95, select F1-maximizing threshold). A 0.75 minimum floor was introduced after discovering that low thresholds caused pervasive false positives on real video frames (oil and egg detected in every frame), exposing the domain gap between clean training images and complex cooking scenes.
- Video pipeline.** A complete keyframe extraction pipeline using PySceneDetect (AdaptiveDetector for scene cuts) + uniform 0.5 FPS sampling + Laplacian blur filtering is now functional. Each keyframe is embedded via the frozen CLIP encoder and classified by the MLP head, producing a per-frame ingredient detection heatmap.
- Nutritional estimation.** Detected ingredients (confirmed by the user with quantities) are matched against an offline USDA FoodData Central SQLite database ($\sim 7,800$ foods) via RapidFuzz fuzzy matching. Results are passed to a locally-hosted **Qwen 2.5 3B** LLM (via Ollama) which generates a structured JSON macro breakdown enforced by a Pydantic schema.

2 Key Metrics Comparison

Metric	Report 1 (K-Means + ViT-B/32)	Current (MLP + ViT-L/14)
Class-level accuracy	98.35%	—
Subclass-level accuracy	85.06%	—
Macro F1 (20 classes)	—	99.2%
Micro F1 (20 classes)	—	99.6%
Multi-label support	No	Yes
Video pipeline	No	Yes
Nutritional output	No	Yes

3 Challenges Addressed

1. **Egg absorption (resolved).** In Report 1, egg (113 images) was absorbed into the oil cluster. The MLP classifier with per-class weighting now correctly identifies eggs with $F1 > 0.96$, as the supervised head learns class-specific decision boundaries rather than relying on global cluster centroids.
2. **False positives on video frames (resolved).** Initial threshold-free inference caused oil and egg to be detected in every video frame. Per-class threshold calibration with a 0.75 floor eliminated these while maintaining 99.2% test F1.
3. **Rice variety indistinguishability (partially resolved).** The ViT-L/14 encoder with trained MLP separates rice varieties better than K-Means on ViT-B/32, though minor confusions between visually similar varieties persist.

[GitHub Repository](#)